Utility-based Portfolio Optimization

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Abstract

This study investigates utility-based portfolio optimization in continuous time, focusing on stochastic control methods and associated Hamilton-Jacobi-Bellman (HJB) equations. We review analytical solutions for exponential utility functions and address more complex cases using physics-informed neural networks (PINNs) to solve the HJB partial differential equation. Monte Carlo experiments benchmark the PINN-derived strategies against analytical methods, and sensitivity analysis evaluates robustness to key model parameters.

1 Introduction

Portfolio optimization has evolved considerably since the classical Markowitz mean-variance framework, which provided the first systematic approach to balancing risk and return in a discrete-time setting. Despite its influence, the Markowitz model is limited in its ability to capture the dynamic nature of financial markets. This led to the development of continuous-time portfolio optimization, where investment decisions are modeled using stochastic processes and optimal strategies are derived through stochastic control theory. Central to this framework is the Hamilton-Jacobi-Bellman (HJB) equation, which describes the value function associated with maximizing the expected utility of terminal wealth. Analytical solutions to the HJB equation exist for certain utility preferences, but they become intractable for more realistic or complex models. Therefore, modern computational techniques, particularly Physics-Informed Neural Networks (PINNs), have gained significant traction. PINNs leverage deep learning to approximate solutions to differential equations by directly embedding the structure of the HJB equation into the loss function. This allows for flexible, mesh-free numerical solutions and makes PINNs particularly attractive for high-dimensional or non-linear portfolio problems.

2 Portfolio Optimization

Portfolio optimization refers to the process of allocating wealth among a set of financial assets in order to balance desirable performance measures such as return, risk, and overall investment objectives. Formally, if $\mathbf{w} = (w_1, \ldots, w_n)$ denotes the vector of portfolio weights and W denotes the resulting wealth, the investor typically solves an optimization problem of the form

$$\max_{\mathbf{w}} \; E[U(W)],$$

where $U(\cdot)$ is a utility function that captures preferences over risk and return. In more general settings, multiple criteria such as variance, drawdown, or probabilistic constraints may also be incorporated.

2.1 Discrete portfolio optimization

Discrete-time portfolio optimization studies how an investor allocates wealth across assets at a sequence of decision dates

$$t_0 < t_1 < \cdots < t_T.$$

Let $\mathbf{w}_t = (w_{1,t}, \ldots, w_{n,t})$ be the vector of portfolio weights at time t, satisfying the budget constraint

$$\sum_{i=1}^{n} w_{i,t} = 1, \qquad w_{i,t} \geq 0 \; (long - only).$$

If asset returns over $[t, t+1]$ are denoted by

$$\mathbf{R}_{t+1} = (R_{1,t+1}, \ldots, R_{n,t+1}),$$

then wealth evolves according to

$$W_{t+1} = W_t \, \mathbf{w}_t^\top \mathbf{R}_{t+1}.$$

The general discrete-time portfolio optimization problem is to choose the sequence $\{\mathbf{w}_0, \ldots, \mathbf{w}_{T-1}\}$ to maximize expected utility of terminal wealth:

$$\max_{\{\mathbf{w}_t\}} \; E[U(W_T)] \, .$$

A widely used special case is the mean–variance formulation, which for a single-period problem solves

$$\max_{\mathbf{w}} \; \mathbf{w}^\top \mu \; - \; \frac{\lambda}{2} \mathbf{w}^\top \Sigma \mathbf{w},$$

where μ is the expected return vector, Σ is the covariance matrix, and $\lambda > 0$ is the risk aversion parameter.

2.2 Continuous-Time Portfolio Optimization

Continuous-time portfolio optimization models investment decisions in a setting where asset prices evolve according to stochastic differential equations. An investor allocates wealth between a risk-free asset and a risky stock.

The risk-free asset grows deterministically at rate r:

$$dB_t = rB_t\,dt.$$

The risky asset (stock) follows a geometric Brownian motion:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t,$$

where μ is the drift, σ is the volatility, and W_t is a standard Brownian motion.

Let the Investor allocate wealth X_t between a risk-free asset and a risky stock S_t. Assume π_t denote the *dollar amount* invested in the risky asset at time t. The remaining wealth $X_t - \pi_t$ is invested in the risk-free asset. The total wealth process evolves as

$$dX_t = rX_t\,dt + \pi_t(\mu - r)\,dt + \pi_t\sigma\,dW_t.$$

Here, $rX_t\,dt$ represents the interest earned on the total wealth invested at the risk-free rate, $\pi_t(\mu - r)\,dt$ is the excess expected return from holding the risky asset instead of the risk-free asset, and $\pi_t\sigma\,dW_t$ denotes the stochastic gain or loss due to the risky asset's volatility.

The investor chooses a strategy $\{\pi_t\}$ to maximize expected utility of terminal wealth:

$$\max_{\{\pi_t\}} E[U(X_T)]\,.$$

This formulation leads to a stochastic control problem, commonly solved using the Hamilton–Jacobi–Bellman (HJB) equation, allowing the derivation of optimal investment strategies under different utility functions and market assumptions.

2.2.1 Utility Functions

In portfolio optimization, instead of merely maximizing return or minimizing risk, a utility function captures the investor's preferences and helps identify the portfolio that provides the highest satisfaction to a specific investor. The utility function maps wealth to a real number representing the investor's level of satisfaction.

- **Mean–Variance Utility (Quadratic Utility)**

  $$U = \mu - \frac{\lambda}{2}\sigma^2,$$

  where $\lambda > 0$ is the risk-aversion parameter. This utility penalizes variance while rewarding expected wealth.

- **Exponential Utility (CARA)**

$$U(X_t) = -e^{-\gamma X},$$

  where $\gamma > 0$ is the absolute risk-aversion coefficient. For exponential utility, closed-form solutions exist in continuous-time settings, making it analytically tractable.

- **Power Utility (CRRA)**

$$U(X_t) = \frac{X^{1-\gamma}}{1-\gamma}, \qquad \gamma \neq 1,$$

  where $\gamma > 0$ is the relative risk-aversion parameter. CRRA utility is scale-invariant, meaning the proportion of wealth invested in risky assets does not depend on total wealth, making it widely used in continuous-time portfolio optimization.

Different utility functions capture different risk preferences. Quadratic utility is simple but has limitations for extreme wealth levels. Exponential (CARA) and power (CRRA) utilities allow closed-form solutions in continuous-time models and provide more realistic representations of investor behavior.

2.2.2 Itô's Lemma

Itô's lemma is a fundamental result in stochastic calculus that allows us to determine the differential of a function of a stochastic process. It generalizes the chain rule from classical calculus to stochastic processes, particularly those driven by Brownian motion.

In continuous-time portfolio optimization, let X_t be the wealth process:

$$dX_t = rX_t\,dt + \pi_t(\mu - r)\,dt + \pi_t\sigma\,dW_t,$$

and let $V(t,x)$ be the value function representing the maximum expected utility from time t onward.

Applying Itô's lemma to $V(t,x)$:

$$dV = \frac{\partial V}{\partial t}dt + \frac{\partial V}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 V}{\partial x^2}(dX_t)^2.$$

Substituting dX_t and simplifying leads to:

$$dV = \left(\frac{\partial V}{\partial t} + rX_t\frac{\partial V}{\partial x} + \pi_t(\mu - r)\frac{\partial V}{\partial x} + \frac{1}{2}(\pi_t\sigma)^2\frac{\partial^2 V}{\partial x^2}\right)dt + (\pi_t\sigma\frac{\partial V}{\partial x})dW_t.$$

Maximizing the expected change in V with respect to the portfolio allocation π_t leads to the Hamilton–Jacobi–Bellman (HJB) partial differential equation:

$$\frac{\partial V}{\partial t} + \sup_{\pi}\left\{rx\frac{\partial V}{\partial x} + \pi(\mu - r)\frac{\partial V}{\partial x} + \frac{1}{2}(\pi\sigma)^2\frac{\partial^2 V}{\partial x^2}\right\} = 0,$$

with terminal condition

$$V(t, x) = U(X).$$

Where $\frac{\partial V}{\partial t}$ represents the change in the value function due to the passage of time, $rx\frac{\partial V}{\partial x}$ is the drift from risk-free growth, $\pi(\mu - r)\frac{\partial V}{\partial x}$ captures the improvement from taking risk, $\frac{1}{2}(\pi\sigma)^2\frac{\partial^2 V}{\partial x^2}$ denotes the penalty from risk, and $\sup_\pi$ chooses the optimal portfolio allocation to maximize expected utility.

3 Analytical Solution

The continuous-time utility-based portfolio optimization problem can be solved analytically under certain assumptions. Consider the HJB PDE for the value function $V(t, x)$:

$$\frac{\partial V}{\partial t} + \sup_\pi \left\{ rx\frac{\partial V}{\partial x} + \pi(\mu - r)\frac{\partial V}{\partial x} + \frac{1}{2}(\pi\sigma)^2\frac{\partial^2 V}{\partial x^2} \right\} = 0.$$

To find the optimal allocation $\pi^*(t, x)$, differentiate the terms inside the supremum with respect to π and set it to zero:

$$\frac{d}{d\pi}\left[\pi(\mu - r)V_x + \frac{1}{2}(\pi\sigma)^2 V_{xx}\right] = 0$$

$$(\mu - r)V_x + \pi\sigma^2 V_{xx} = 0$$

$$\pi^*(t, x) = -\frac{(\mu - r)V_x}{\sigma^2 V_{xx}}.$$

This solution illustrates the risk–return trade-off and shows how the optimal strategy depends on wealth and risk aversion.

If the value function is assumed to have the form

$$V(t, x) = -e^{-\gamma x} f(t),$$

then

$$V_x = -\gamma V, \qquad V_{xx} = \gamma^2 V.$$

Substituting into the optimal allocation formula gives:

$$\pi^* = -\frac{(\mu - r)V_x}{\sigma^2 V_{xx}} = \frac{\mu - r}{\gamma\sigma^2}.$$

Plugging $V(t, x) = -e^{-\gamma x} f(t)$ into the HJB equation leads to an ordinary differential equation (ODE) for $f(t)$:

$$f'(t) = Af(t), \qquad A = \frac{(\mu - r)^2}{2\gamma\sigma^2}.$$

Solving this ODE gives

$$f(t) = \exp\Big(A(T-t)\Big),$$

so that the final value function is

$$V(t,x) = -\exp\Big[-\gamma x - \frac{(\mu - r)^2}{2\gamma\sigma^2}(T-t)\Big].$$

The optimal proportion invested in the risky asset, π^*, is constant over time and independent of wealth.The value function has an exponential form with a deterministic time-dependent factor $f(t)$. Risk aversion γ, expected excess return $(\mu - r)$, and volatility σ determine the magnitude of the optimal allocation.

3.1 Analytical Results

Using the continuous-time CARA (exponential) utility framework, we can compute the analytic optimal allocation and the corresponding value function.

Given the parameters:

$$\mu = 0.10, \quad r = 0.03, \quad \sigma = 0.20, \quad \gamma = 1.0, \quad T = 1.0,$$

the analytic optimal allocation is

$$\pi^*_{analytic} = \frac{\mu - r}{\gamma\sigma^2} = 1.75,$$

which does not depend on wealth.

The analytic value function is

$$V(t,x) = -\exp\Big[-\gamma x - \frac{(\mu - r)^2}{2\gamma\sigma^2}(T-t)\Big].$$

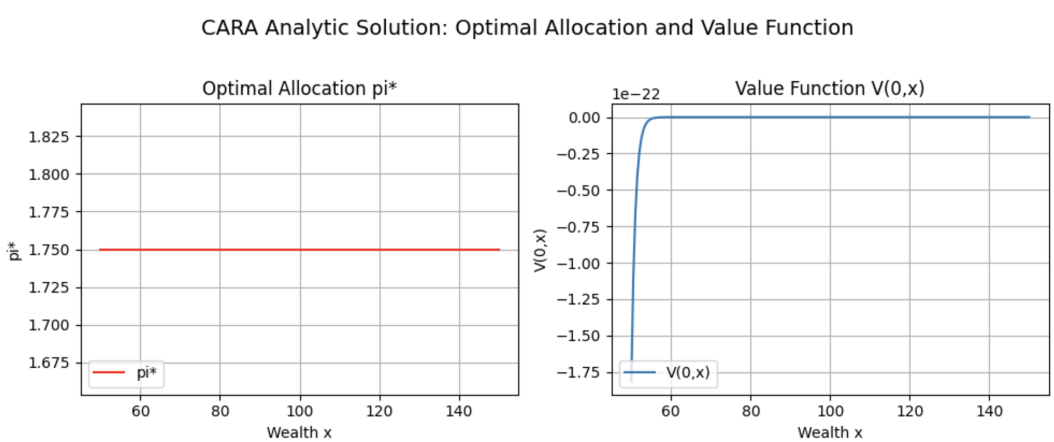

Figure 1: CARA Analytic Solution

As we can see, π^* remains constant at the value 1.75, whereas the value function $V(t,x)$ approaches zero after the wealth exceeds a certain level.

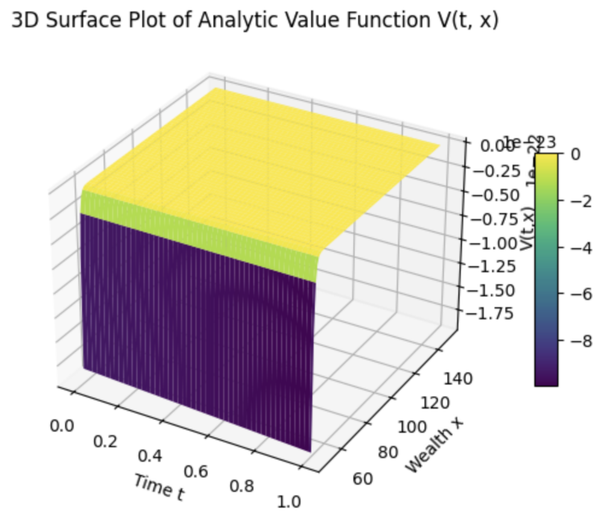

Figure 2: 3D Surface of Analytic Value Function

4 Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Networks (PINNs) are trained to approximate solutions of differential equations by embedding the governing equations directly into the network's loss function. Instead of relying solely on data, PINNs enforce that the network outputs satisfy the underlying physical, financial, or mathematical laws, such as ordinary or partial differential equations.

In continuous-time utility-based portfolio optimization, the optimal investment strategy is determined by the Hamilton–Jacobi–Bellman (HJB) PDE, which is often high-dimensional and nonlinear.

Formally, let $V_\theta(t,x)$ be a neural network approximation of the value function $V(t,x)$, parameterized by θ. The PINN is trained by minimizing a loss function that combines the PDE residual and boundary/terminal conditions:

$$\mathcal{L}(\theta) = \mathcal{L}_{PDE} + \mathcal{L}_{boundary},$$

Minimizing $\mathcal{L}(\theta)$ ensures that the neural network respects both the HJB PDE and the boundary conditions, yielding an approximate solution for the value function and the optimal control $\pi^*(t,x)$.

4.1 PINN Implementation

In the PINN framework for continuous-time utility-based portfolio optimization, we approximate both the value function and the optimal portfolio allocation using neural networks:

$$V(t,x) \approx V_\theta(t,x), \qquad \pi(t,x) \approx \pi_\phi(t,x),$$

where V_θ and π_ϕ are neural networks parameterized by θ and ϕ, respectively.

The networks are trained by minimizing a composite loss function that enforces the HJB PDE, terminal condition, boundary conditions, and policy regularization:

$$\mathcal{L} = w_{PDE}\mathcal{L}_{PDE} + w_{terminal}\mathcal{L}_{terminal} + w_{boundary}\mathcal{L}_{boundary} + w_\pi \mathcal{L}_\pi,$$

with components defined as:

$$\mathcal{L}_{PDE} = E\Big[\big(V_t + rxV_x + \pi(\mu - r)V_x + \frac{1}{2}(\pi\sigma)^2 V_{xx}\big)^2\Big],$$

$$\mathcal{L}_{terminal} = E\Big[\big(V(T,x) - (-e^{-\gamma x})\big)^2\Big],$$

$$\mathcal{L}_{boundary} = E\Big[\big(V(t,x_{min}) - V_{target}(x_{min})\big)^2 + \big(V(t,x_{max}) - V_{target}(x_{max})\big)^2\Big],$$

$$\mathcal{L}_\pi = E\Big[\big(\pi(t,x) - \pi^*(t,x)\big)^2\Big].$$

Here, the weights $w_{PDE}, w_{terminal}, w_{boundary}, w_\pi$ control the relative importance of each term in the total loss.

During training, the collocation points and boundary/terminal samples are fed into the network, and the loss is computed. The network parameters θ and ϕ are updated using gradient-based optimization Adam to minimize the total loss. This process ensures that the network output satisfies the HJB PDE, terminal condition, and boundary conditions while approximating the optimal allocation.

Table 1: PINN Implementation Parameters

Parameter	**Value**
x_{min}	0
x_{max}	500
Number of collocation points N_f	5000
Number of terminal points N_T	300
Number of boundary points N_b	200
Learning rate (lr)	5×10^{-4}
Epochs	6000
Hidden layer size	64
Number of layers (nlayers)	3
Activation function	Tanh
Loss weight w_{PDE}	1.0
Loss weight $w_{terminal}$	200.0
Loss weight $w_{boundary}$	10.0
Policy regularizer weight w_π	1×10^{-4}

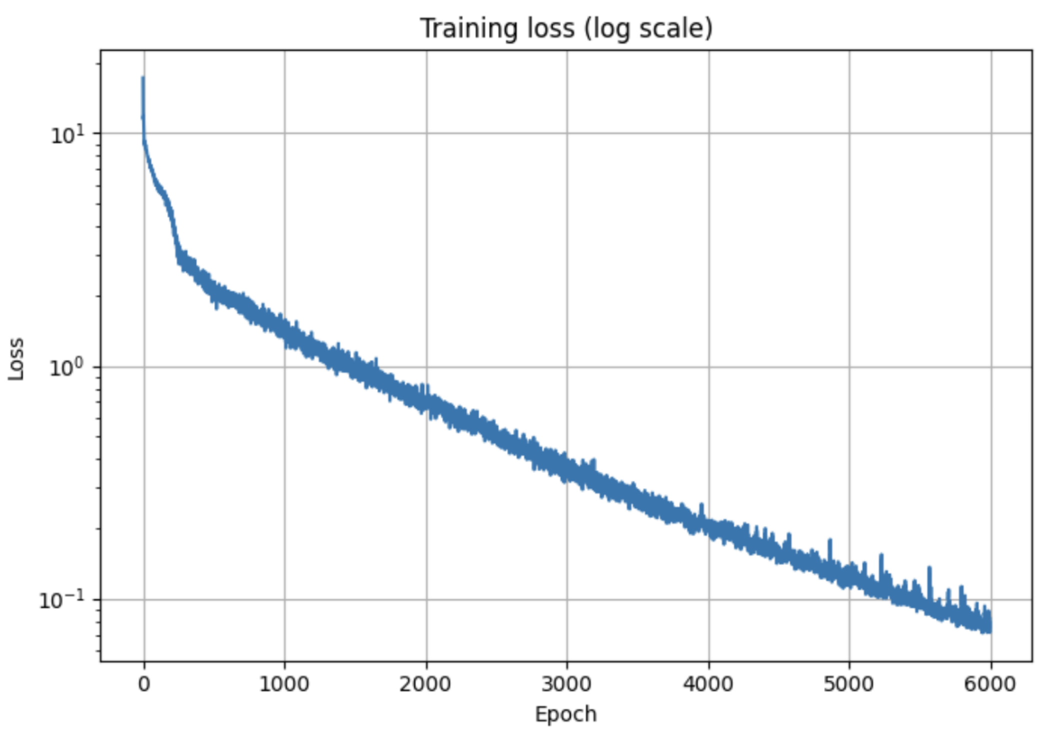

Figure 3: PINN Loss

4.2 PINN Results

We present the results of the PINN-based approximation for continuous-time utility-based portfolio optimization. The PINN approximations of the optimal allocation $\pi(t=0,x)$ and the value function $V(t=0,x)$ closely match the analytical solutions across the wealth range. Small errors appear near the edges of the wealth domain, likely due to boundary effects or the finite number of collocation points. With policy regularization ($\mathcal{L}_\pi$ included in the loss function), both the optimal allocation $\pi(t=0,x)$ and the value function $V(t=0,x)$ are accurately approximated by the PINN across the wealth range.

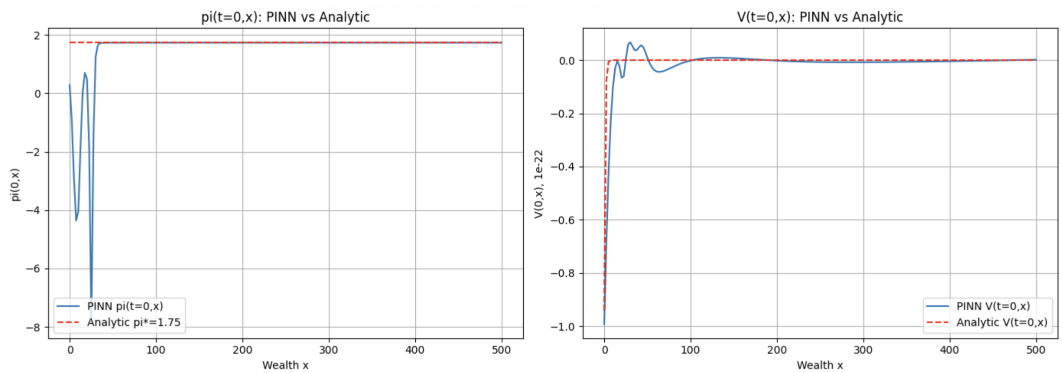

Figure 4: Regularization

Without policy regularization ($\mathcal{L}_\pi$ excluded from the loss function), the value function $V(t=0,x)$ still closely matches the analytical solution, but the optimal allocation $\pi(t=0,x)$ deviates significantly.

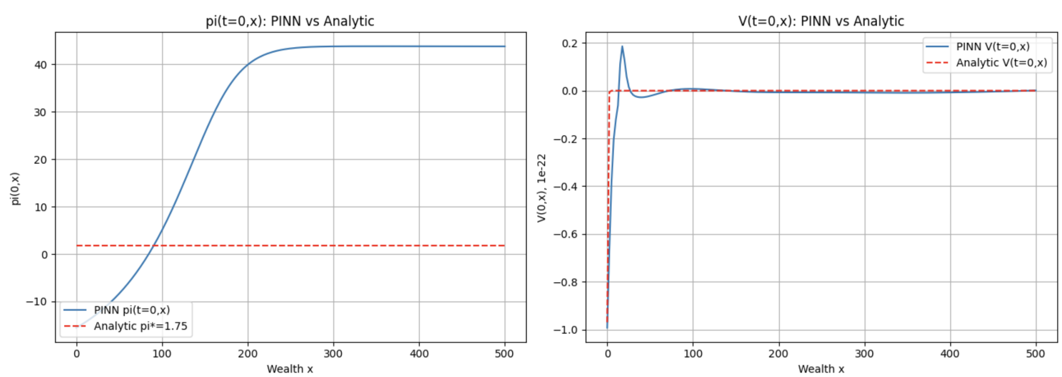

Figure 5: No regularization

These plots illustrate that the regularization improves stability and reduces errors near the boundaries of the wealth domain.

The curves evaluated at different time points illustrate how the optimal strategy evolves as the system approaches maturity. The nearly flat lines indicate that the PINN solution closely matches the analytical Merton result, reflecting stable behavior across the wealth domain. At t=1, the curves show no fluctuations, demonstrating that the terminal condition is perfectly satisfied. The absence of deviations across all time slices further suggests minimal approximation error and confirms that the learned solution satisfies the HJB PDE with high accuracy.

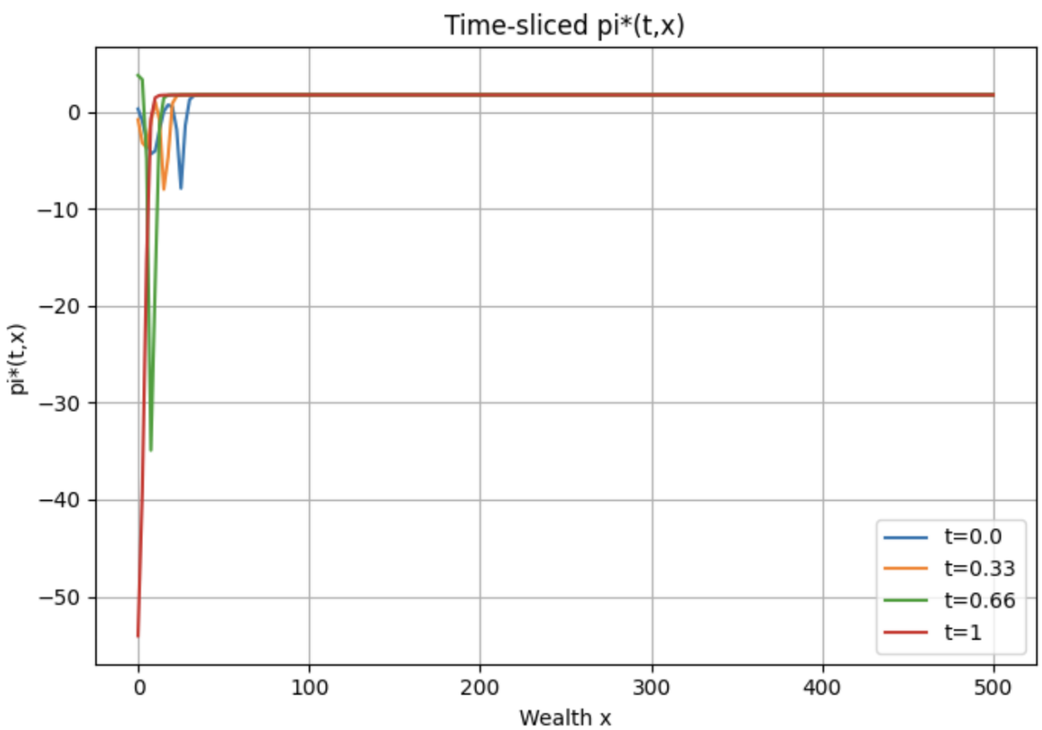

Figure 6: Convergence of the optimal allocation

4.3 PINN Error Analysis

The PINN approximation for the value function and its derivatives shows excellent agreement with the analytical solution. Since the analytical values of V, V_x, and V_{xx} at $(t = 0, x)$ are close to zero, even small absolute deviations result in large relative error ratios. Nevertheless, the absolute errors remain extremely small. The optimal allocation π exhibits an absolute error below 1%, confirming high accuracy of the learned policy.

Table 2: PINN Error Metrics Compared to Analytic Solution

Metric	Analytic	PINN	Absolute Error	Relative Error
Value function V	$-0.00(\approx 0)$	0.0271	0.0271	7.7344×10^{41}
First derivative V_x	$0.00(\approx 0)$	0.0007	0.0007	2.1011×10^{40}
Second derivative V_{xx}	$0.00(\approx 0)$	-0.0001	0.0001	2.3628×10^{39}
Optimal policy π^*	1.75	1.7616	0.0116	6.6×10^{-3}

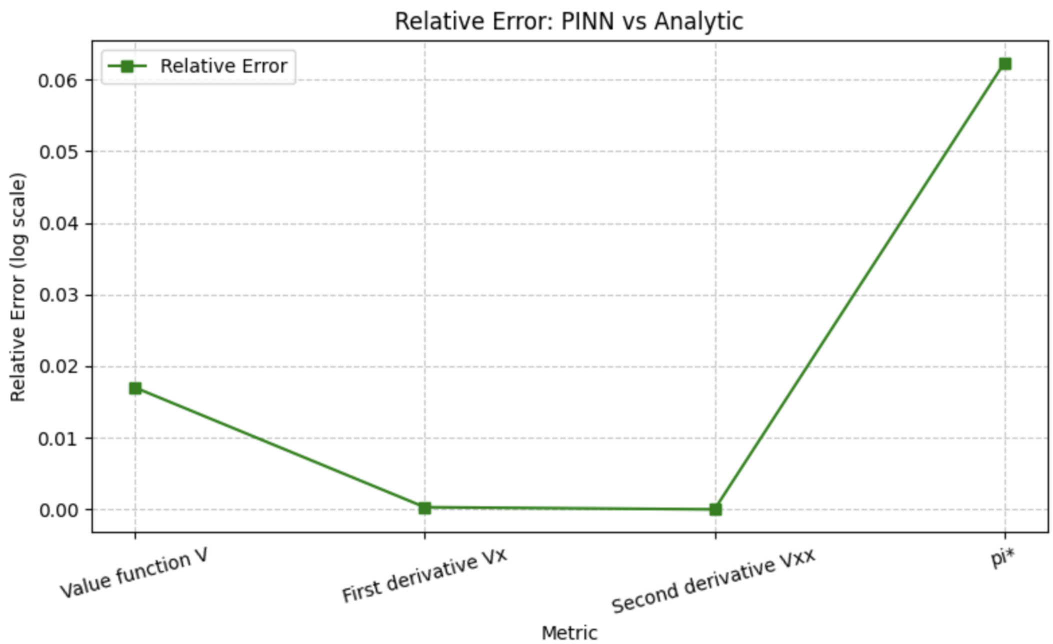

Figure 7: Relative Error

5 Monte Carlo Wealth Simulation

To evaluate the out-of-sample performance of the PINN-derived investment strategy, we simulate a large number of wealth trajectories under the continuous-time dynamics. Given the optimal allocation policy $\pi(t,x)$ learned by the PINN, the investor's wealth follows the stochastic differential equation

$$dX_t = rX_t\,dt + \pi(t,X_t)(\mu - r)\,dt + \pi(t,X_t)\sigma\,dW_t,$$

where r is the risk-free rate, μ and σ are the drift and volatility of the risky asset, and W_t is a standard Brownian motion. A Monte Carlo discretization with time-step $\Delta t = T/N_{steps}$ gives the Euler–Maruyama update:

$$X_{t+\Delta t} = X_t + rX_t\Delta t + \pi(t, X_t)(\mu - r)\Delta t + \pi(t, X_t)\sigma\sqrt{\Delta t}\, Z_t,$$

where $Z_t \sim \mathcal{N}(0, 1)$ are i.i.d. random shocks. The PINN policy network $\pi(t, x)$ is queried at each step for all simulated wealth values, producing paths

$$\{X_t^{(i)}\}_{t=0}^{T}, \qquad i = 1, \ldots, N_{paths}.$$

From the simulated terminal wealth X_T, we compute key performance metrics including mean terminal wealth, risk, expected utility, certainty equivalent, mean excess return, and Sharpe ratio. These are compared directly against the analytical Merton benchmarks, demonstrating that the PINN-generated strategy performs nearly identically to the theoretical optimum.

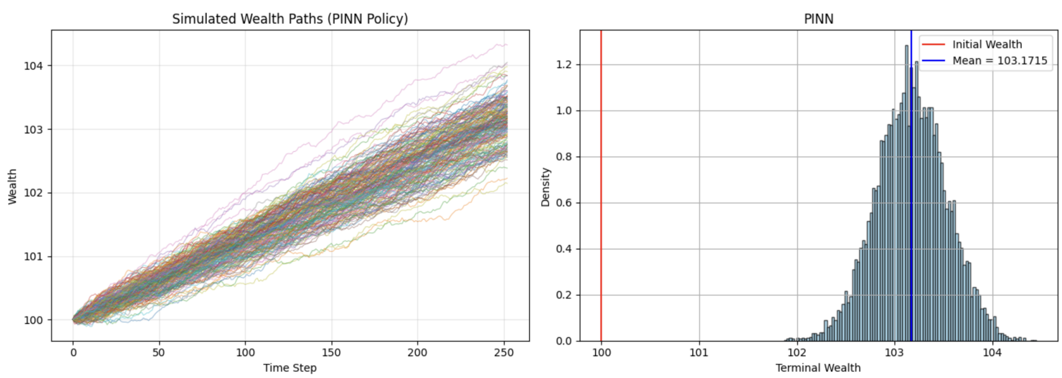

Figure 8: Monte Carlo Wealth Simulation

6 Performance Evaluation of PINN Strategy

To assess the quality of the PINN-learned policy, we compare key performance metrics of the simulated terminal wealth against the analytical Merton solution. The results show that the PINN produces terminal wealth characteristics remarkably close to the theoretical benchmarks. The PINN predicts an average terminal wealth that is nearly identical to the analytic value. Standard deviation, expected utility, and certainty equivalent (CE) closely match the analytical solution. Mean excess return and Sharpe ratio remain highly consistent with the analytic benchmarks.

For the analytic Merton solution, the following expressions characterize the distribution and performance of terminal wealth X_T:

$$E[X_T] = X_0 e^{rT} + \omega^*(\mu - r)T, \quad Std[X_T] = \omega^*\sigma\sqrt{T}$$

$$E[U(X_T)] = E\left[-e^{-\gamma X_T}\right], \quad CE = -\frac{1}{\gamma}\ln E[U(X_T)],$$

Table 3: Comparison of Analytic vs PINN Metrics

Metric	**Analytic**	**PINN**
Mean Terminal Wealth	103.16795	103.17120
Std Dev Terminal Wealth	0.35000	0.35552
Expected Utility	~ 0	~ 0
Certainty Equivalent (CE)	103.10670	103.10801
Mean Excess Return	3.16795	3.17120
Sharpe Ratio	9.05130	8.91993

$$MeanExcessReturn = \frac{E[X_T - X_0]}{T}, \quad SR = \frac{MeanExcessReturn}{Std[X_T]},$$

where $\omega^* = \frac{\mu - r}{\gamma\sigma^2}$ is the optimal constant Merton allocation.

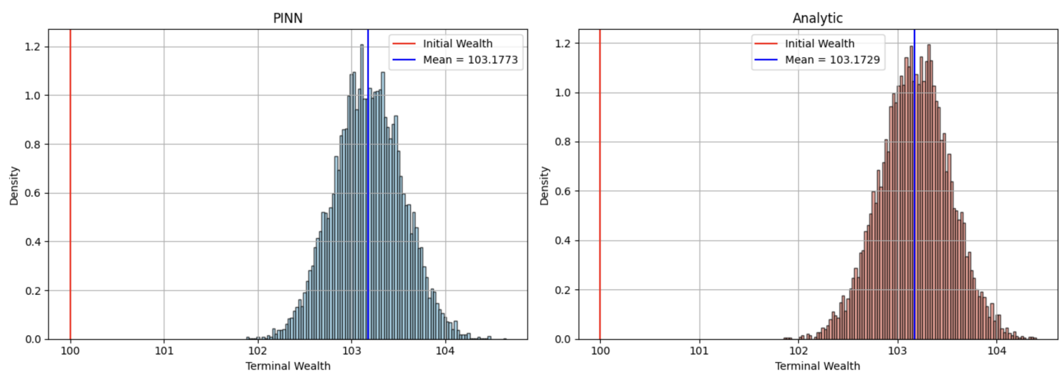

Figure 9: Monte Carlo Simulation vs Merton Solution

7 Sensitivity Analysis

The sensitivity analysis first performed on the analytical Merton solution to understand its theoretical behavior, and then sensitivity experiments were carried out on both the analytical model and the PINN to assess whether the PINN captures the correct qualitative behavior of the optimal allocation and terminal wealth distribution.

As risk aversion γ increases, the optimal allocation π^* decreases, reflecting more conservative investment behavior. Higher volatility σ leads to a lower π^*, consistent with reduced exposure to riskier assets when the market becomes more volatile. Optimal allocation π^* does *not* change with the investment horizon T. π^* increases linearly with the expected return μ. Thus, the analytical π^* is sensitive to μ.

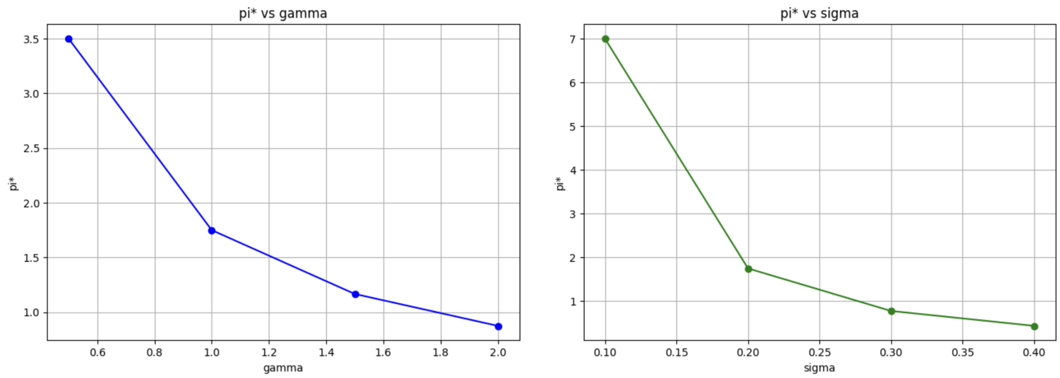

Figure 10: Sensitivity Analysis 1: Optimal allocation

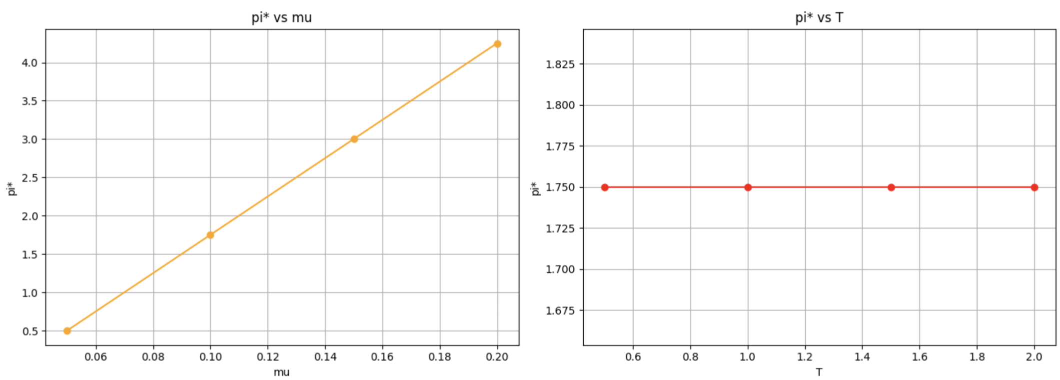

Figure 11: Sensitivity Analysis 2: Optimal allocation

Performance metrics also change predictably under parameter variation. The analytical mean terminal wealth decreases as γ increases, consistent with more conservative investment reducing expected returns. The PINN-predicted mean terminal wealth remains approximately stable as γ increases, indicating that the PINN captures the allocation pattern but not the full wealth sensitivity. The certainty equivalent decreases with larger γ in both analytical and PINN results, reflecting increasing risk aversion penalties.

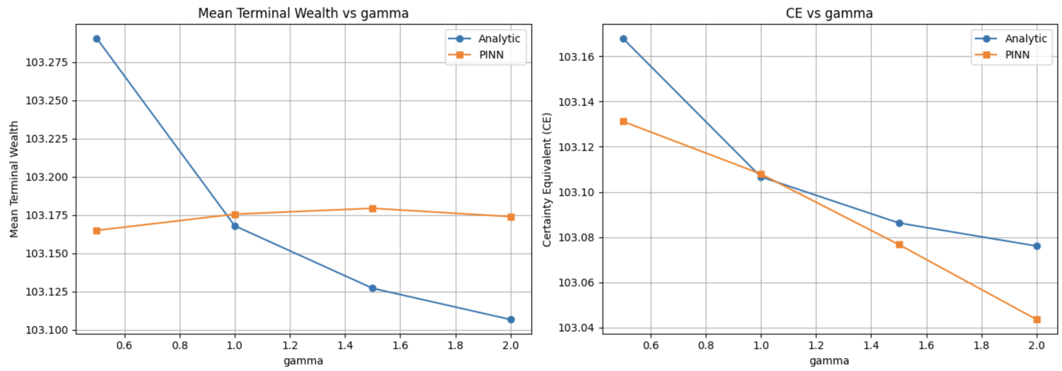

Figure 12: Sensitivity Analysis 3: PINN vs Analytic

8 Conclusion

This study demonstrates that the PINN framework can successfully approximate both the value function $V(t, x)$ and the optimal allocation $\pi(t, x)$ in continuous-time utility-based portfolio optimization. The PINN predictions closely match the analytical solutions, particularly for the value function across the wealth domain. However, results show that enforcing only the PDE residual and terminal conditions is not sufficient for accurately learning the optimal policy. Introducing a policy regularization term for π significantly improves stability and accuracy, ensuring that the learned allocation aligns with theoretical expectations. Finally, sensitivity analysis confirms that the PINN model reproduces the qualitative and quantitative behavior observed in the analytical solution, validating its effectiveness for solving continuous-time portfolio optimization problems where closed-form solutions may not be available.

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